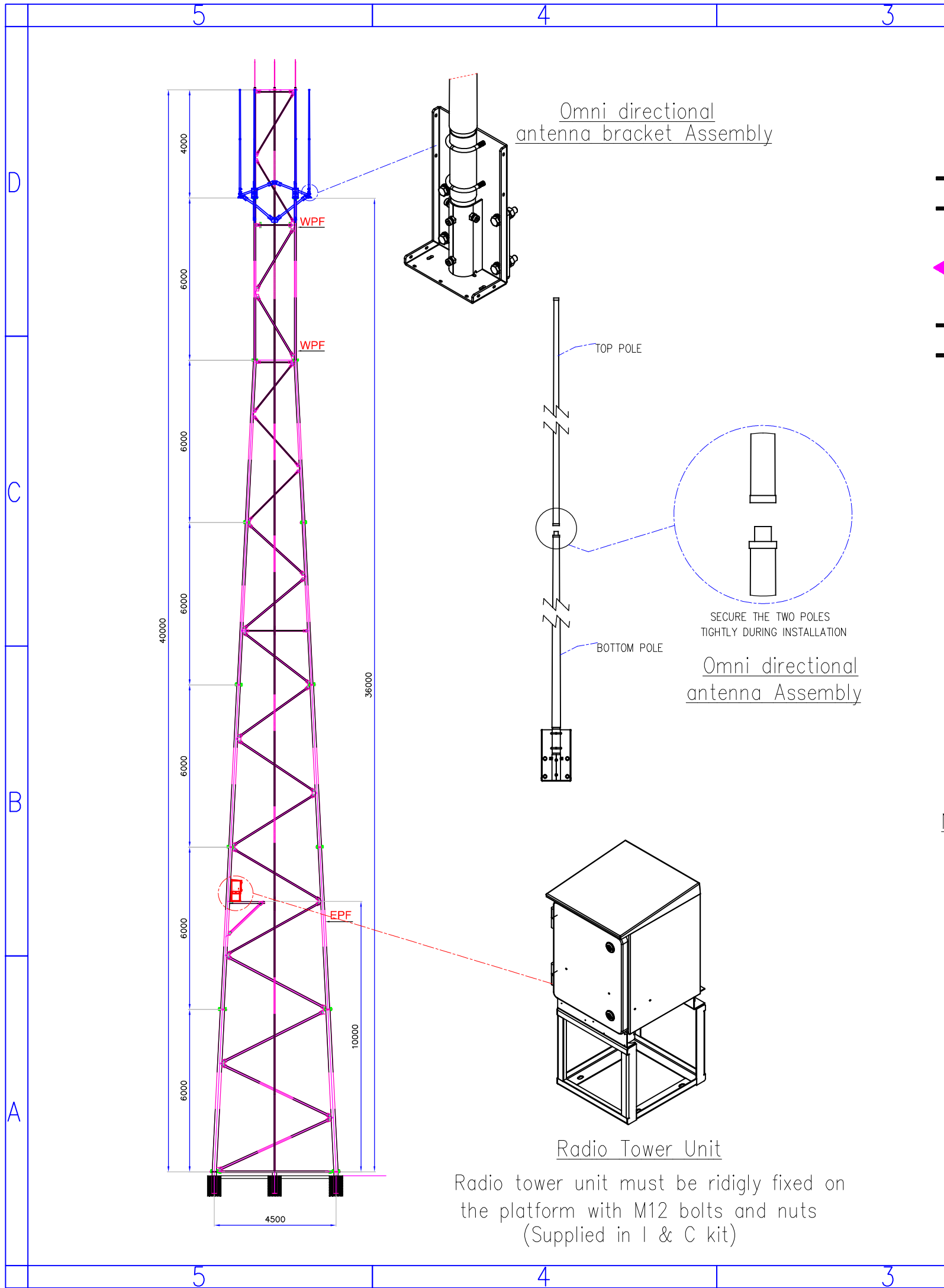
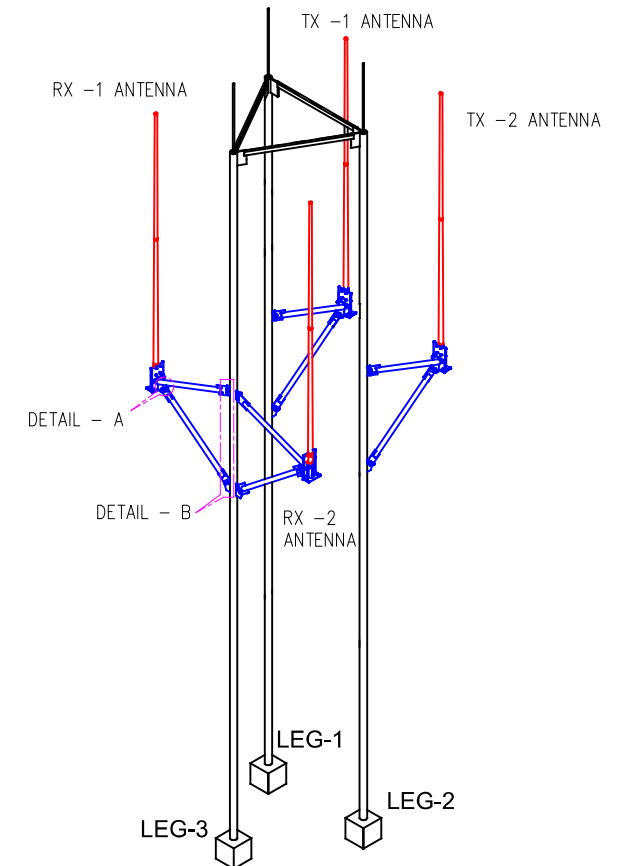
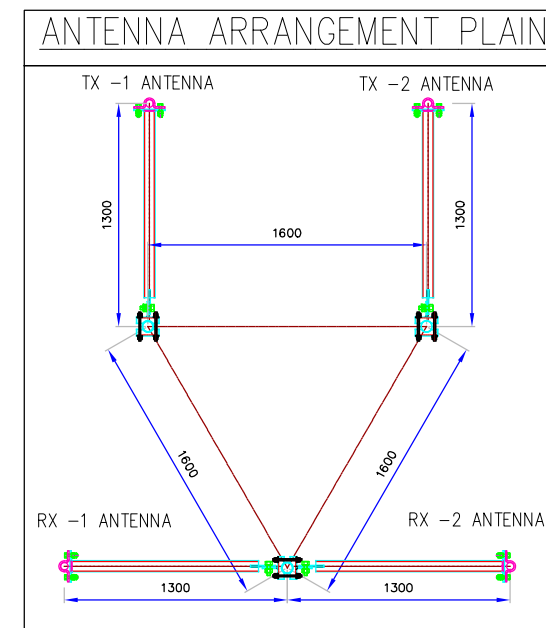
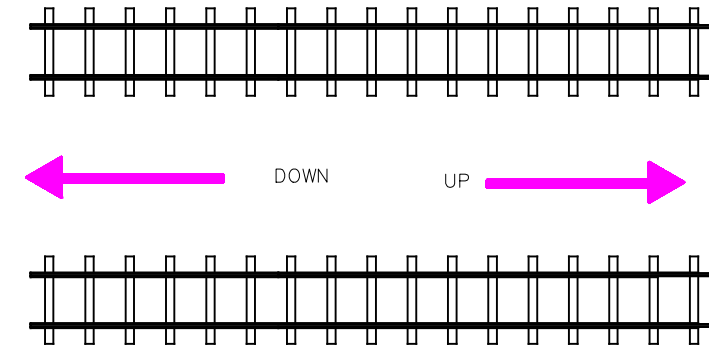




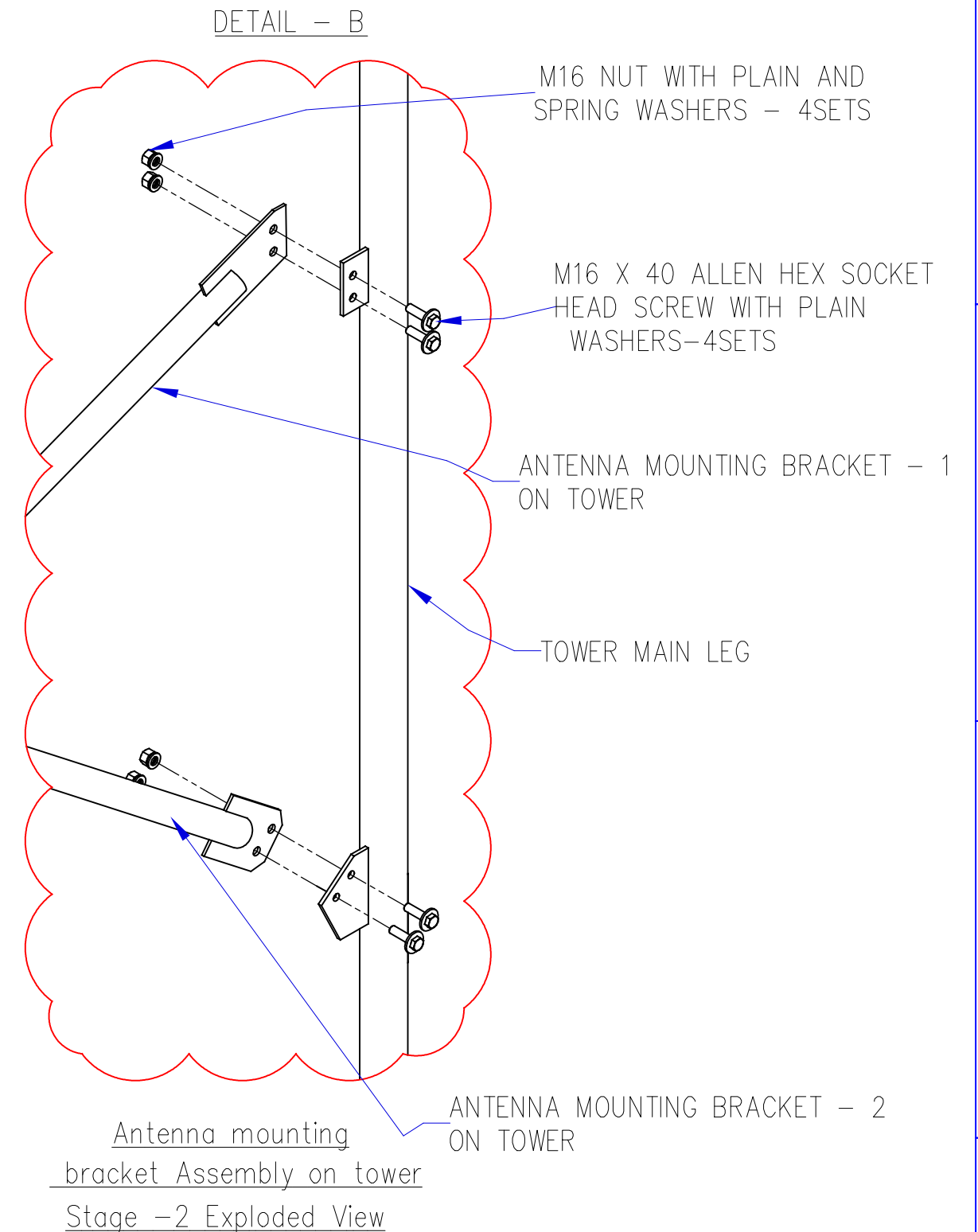
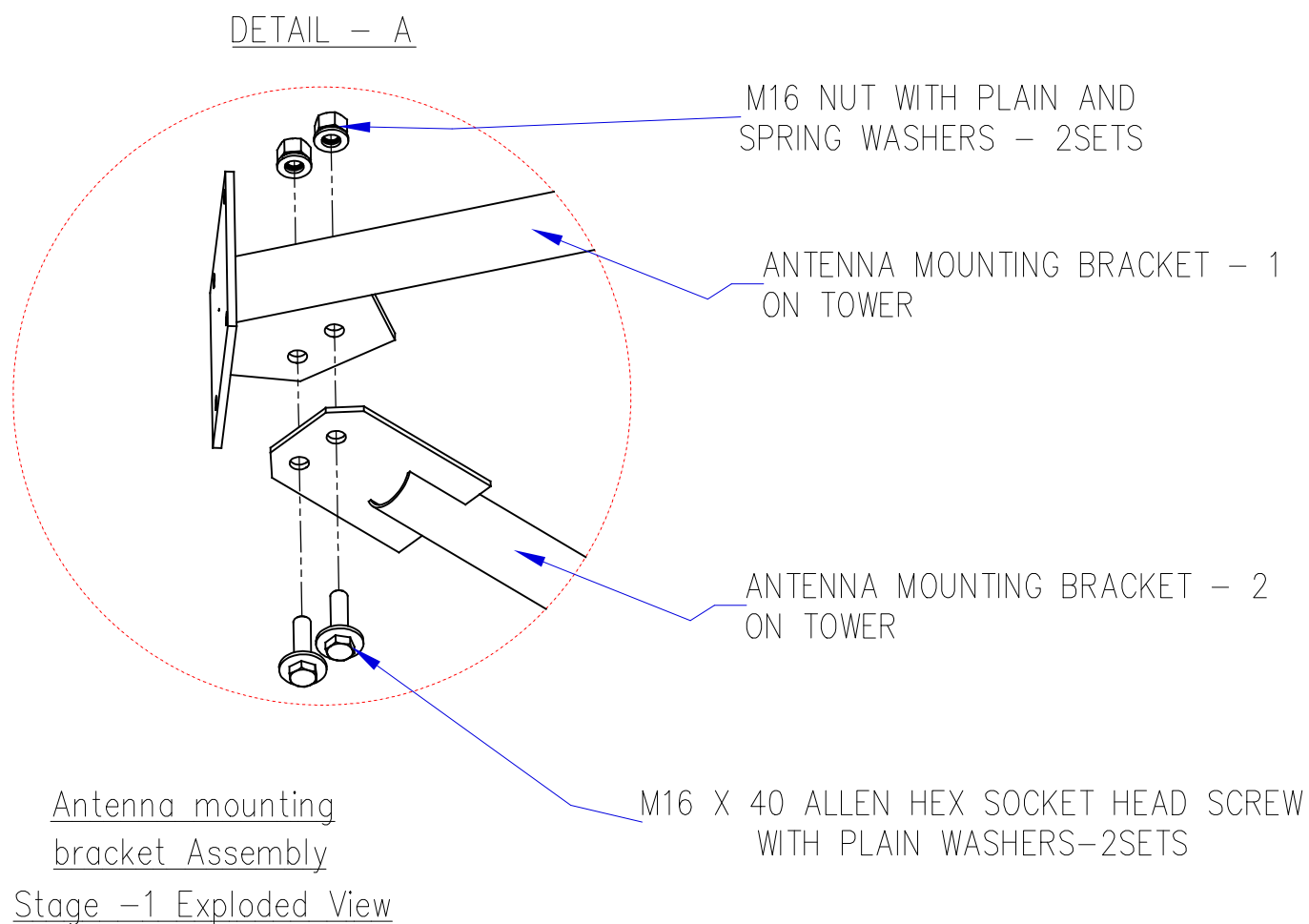
REV.	CHANGE DETAILS	DATE	PREPARED	APPD.
1.1	Updated Antenna mounting structure	20.09.25	SIVAJI	YS
1.2	Added torque details	25.09.25	SIVAJI	YS



Notes:

- Antennas must be installed at the height indicated in the drawing (Top of the antenna must be below the lightning arrester)
- Orientation of Tx & Rx antennas must be maintained as shown in this drawing without fail.
- Weather - Proofing must be done for all RF cable joints
- All Fasteners shall be of SS304 grade material

PROJECT: KAVACH		HBL HBL Engineering Ltd.		NAME	DESIGNATION	SIGN.
ASSEMBLY: STATIONARY KAVACH			DESIGN	CH SIVAJI GANESH	Sr. ENGINEER	CH. Sivaiganesh CH. Sivaiganesh (Sep 25, 2023 09:44:19 GMT+5.5)
			CHECKED	J V V PRASAD	Sr. MANAGER	J V V Prasad
			APPROVED	Y SUBRAHMANYAM	GENERAL MANAGER (ENGINEERING)	Subrahmanyam Y Subrahmanyam Y (Sep 25, 2023 09:56:59 GMT+5.5)
SCALE: NTS	SIZE: A3	TITLE: Installation arrangement for 10.2dBi omni — directional antennas & Radio Tower Unit	DRAWING NO. 5 16 67 0983			REV.
	SHEET: 1 OF 3		PART NO.			1.2
EFFECTIVE FROM:			TOLERANCE NOT SPECIFIED SHALL BE AS PER IS : 2102 — 1993.			
ALL DIMENSIONS ARE IN mm		THIS DRAWING IS THE PROPERTY OF HBL POWER SYSTEMS LIMITED AND MUST NOT BE COPIED, REPRODUCED, TRANSMITTED OR DISCLOSED TO ANY THIRD PARTY WITHOUT THE WRITTEN CONSENT OF THE COMPANY.				

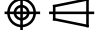


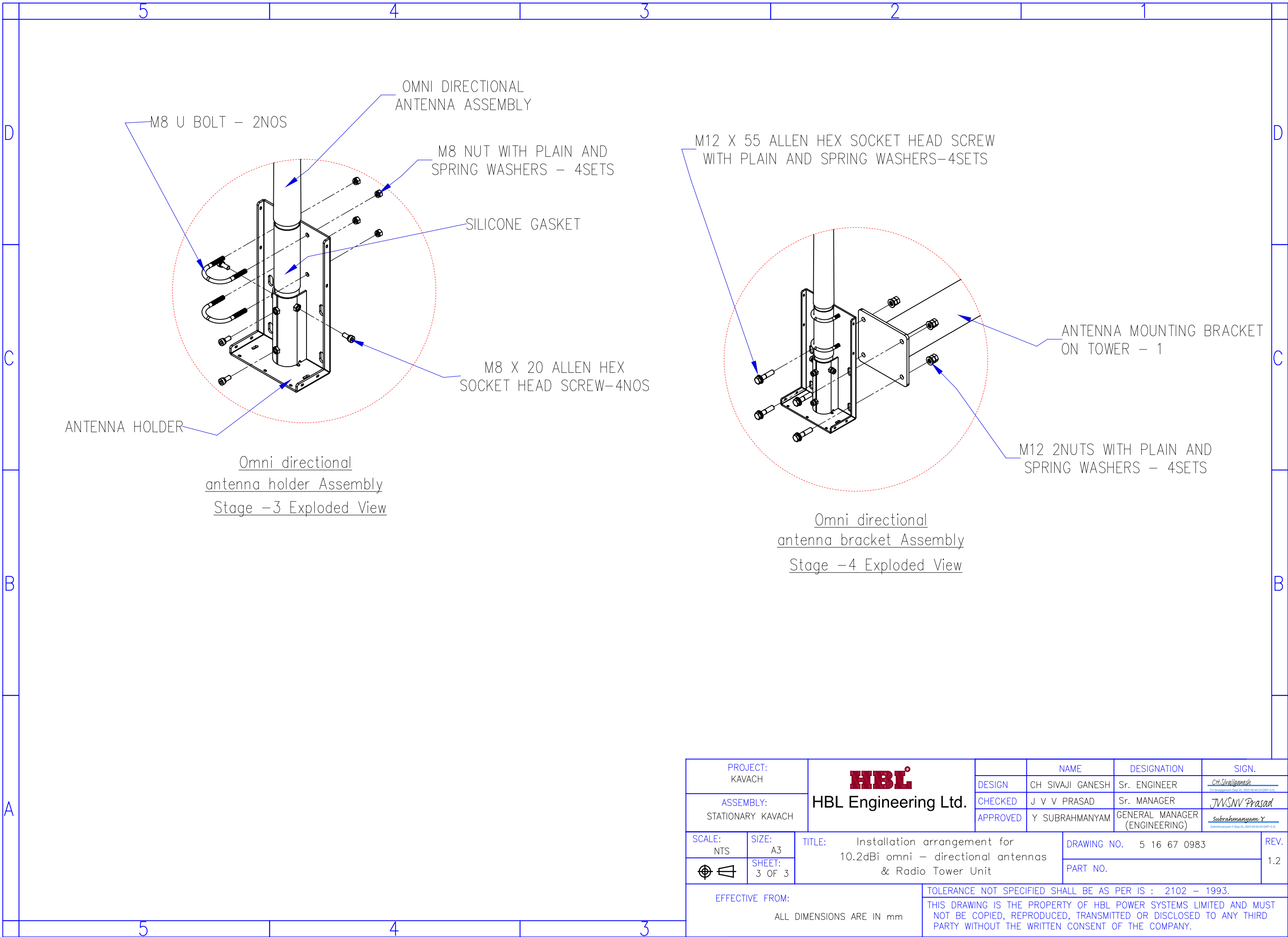
NOTE:

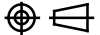
1. All bolts & nuts shall be tightened with torque spanner, with torque set as shown below.

Torque (Nm) - SS304 (8.8/9.8 grade)	
Size	Torque(Nm)
M8	25
M10	50
M12	85
M16	215

2. Paint mark shall be applied on nuts after torque tightening.

PROJECT: KAVACH		 HBL Engineering Ltd.		NAME	DESIGNATION	SIGN.
ASSEMBLY: STATIONARY KAVACH			DESIGN	CH SIVAJI GANESH	Sr. ENGINEER	<u>CH.Sivajiganesh</u> CH.Sivajiganesh 1 Day 25, 2022 09:56:03 (GMT+5:30)
			CHECKED	J V V PRASAD	Sr. MANAGER	<u>J V V V Prasad</u> J V V V Prasad 1 Day 25, 2022 09:56:03 (GMT+5:30)
			APPROVED	Y SUBRAHMANYAM	GENERAL MANAGER (ENGINEERING)	<u>Subrahmanyam Y</u> Subrahmanyam Y 1 Day 25, 2022 09:56:03 (GMT+5:30)
SCALE: NTS	SIZE: A3	TITLE: Installation arrangement for 10.2dBi omni — directional antennas & Radio Tower Unit	DRAWING NO. 5 16 67 0983			REV.
	SHEET: 2 OF 3		PART NO.			1.2
EFFECTIVE FROM:			TOLERANCE NOT SPECIFIED SHALL BE AS PER IS : 2102 – 1993.			
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PROJECT: KAVACH		HBL HBL Engineering Ltd.	NAME	DESIGNATION	SIGN.
ASSEMBLY: STATIONARY KAVACH			DESIGN	CH SIVAJI GANESH	CH Sivajiganesh CH Sivajiganesh (Dep-25, 0025, 09-04-19 EMT+1-5)
			CHECKED	J V V PRASAD	J V V Prasad J V V Prasad (Dep-25, 0025, 09-04-19 EMT+1-5)
			APPROVED	Y SUBRAHMANYAM	Y Subrahmanyam Y Subrahmanyam Y (Dep-25, 0025, 09-04-19 EMT+1-5)
SCALE: NTS	SIZE: A3	TITLE: Installation arrangement for 10.2dBi omni – directional antennas & Radio Tower Unit	DRAWING NO. 5 16 67 0983		REV.
	SHEET: 3 OF 3		PART NO.		1.2
EFFECTIVE FROM:			TOLERANCE NOT SPECIFIED SHALL BE AS PER IS : 2102 – 1993.		
ALL DIMENSIONS ARE IN mm			THIS DRAWING IS THE PROPERTY OF HBL POWER SYSTEMS LIMITED AND MUST NOT BE COPIED, REPRODUCED, TRANSMITTED OR DISCLOSED TO ANY THIRD PARTY WITHOUT THE WRITTEN CONSENT OF THE COMPANY.		

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Signature: Subrahmanyam Y
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Title: GM - Engineering

Installation arrangement for 10.2dBi omni - directional antennas & Radio Tower Unit

Final Audit Report

2025-09-25

Created:	2025-09-25 (India Standard Time)
By:	Subrahmanyam Y (subrahmanyam.yadavalli@hbl.in)
Status:	Signed
Transaction ID:	CBJCHBCAABAA-jkfsVIKfvMv-maT2GzBcuy3bcx3NB4x

"Installation arrangement for 10.2dBi omni - directional antennas & Radio Tower Unit" History

-  Document created by Subrahmanyam Y (subrahmanyam.yadavalli@hbl.in)
2025-09-25 - 9:36:34 AM GMT+5.5- IP address: 122.187.81.114
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 Agreement completed.

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